WEEK 9 – HOUSE PRICE ANALYSIS USING POWER BI

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1. Introduction

House Price Analysis is a data analytics project developed using Microsoft Power BI. The project focuses on analysing house prices based on various property features such as bedrooms, bathrooms, living area, location, condition, renovation status and year of construction.

The dataset used in this project is the King County House Sales Dataset (kc_house_data.csv). Power BI is used to clean, transform, analyse and visualize the data through an interactive dashboard.

2. Objectives

- To import and analyse housing data using Power BI.
- To clean the dataset and handle invalid or missing values.
- To create calculated columns for better analysis.
- To categorize properties based on price and living area.
- To compare house prices using charts and visualizations.

- To develop an interactive dashboard for understanding housing trends.

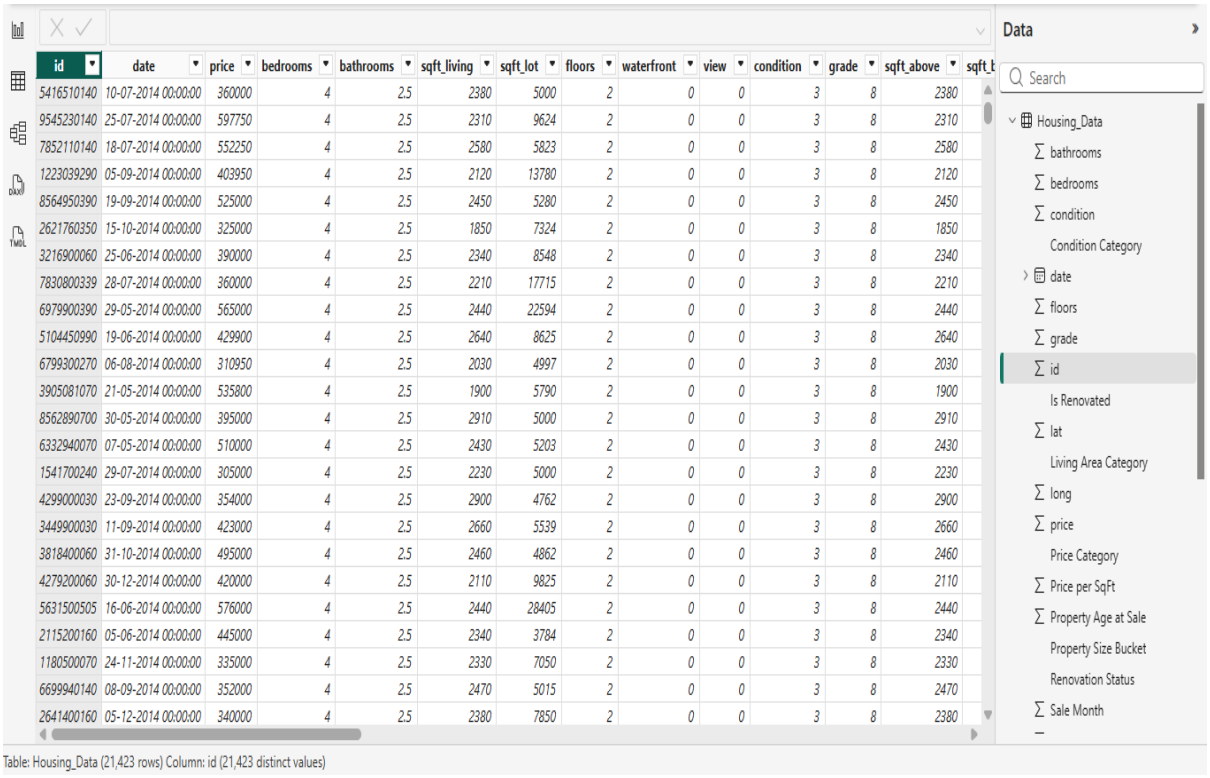
3. Dataset Description

The project uses the kc_house_data.csv dataset.

The dataset contains information about houses, including:

- ID – Unique property identification number.
- Date – Date of house sale.
- Price – Selling price of the house.
- Bedrooms – Number of bedrooms.
- Bathrooms – Number of bathrooms.
- Sqft Living – Living area of the house.
- Sqft Lot – Total land area.
- Floors – Number of floors.
- Yr Built – Year of construction.
- Yr Renovated – Year of renovation.
- Condition – Overall condition of the property.
- Waterfront – Indicates whether the property has a waterfront view.
- Latitude and Longitude – Geographic location details.

4. Data Import and Cleaning



id	date	price	bedrooms	bathrooms	sqft_living	sqft_lot	floors	waterfront	view	condition	grade	sqft_above	sqft_l
5416510140	10-07-2014 00:00:00	360000	4	2.5	2380	5000	2	0	0	3	8	2380	
9545230140	25-07-2014 00:00:00	597750	4	2.5	2310	9624	2	0	0	3	8	2310	
7852110140	18-07-2014 00:00:00	552250	4	2.5	2580	5823	2	0	0	3	8	2580	
1223039290	05-09-2014 00:00:00	403950	4	2.5	2120	13780	2	0	0	3	8	2120	
8564950390	19-09-2014 00:00:00	525000	4	2.5	2450	5280	2	0	0	3	8	2450	
2621760350	15-10-2014 00:00:00	325000	4	2.5	1850	7324	2	0	0	3	8	1850	
3216900060	25-06-2014 00:00:00	390000	4	2.5	2340	8548	2	0	0	3	8	2340	
7830800339	28-07-2014 00:00:00	360000	4	2.5	2210	17715	2	0	0	3	8	2210	
6979900390	29-05-2014 00:00:00	565000	4	2.5	2440	22594	2	0	0	3	8	2440	
5104450990	19-06-2014 00:00:00	429900	4	2.5	2640	8625	2	0	0	3	8	2640	
6799300270	06-08-2014 00:00:00	310950	4	2.5	2030	4997	2	0	0	3	8	2030	
3905081070	21-05-2014 00:00:00	535800	4	2.5	1900	5790	2	0	0	3	8	1900	
8562890700	30-05-2014 00:00:00	395000	4	2.5	2910	5000	2	0	0	3	8	2910	
6332940070	07-05-2014 00:00:00	510000	4	2.5	2430	5203	2	0	0	3	8	2430	
1541700240	29-07-2014 00:00:00	305000	4	2.5	2230	5000	2	0	0	3	8	2230	
4299000030	23-09-2014 00:00:00	354000	4	2.5	2900	4762	2	0	0	3	8	2900	
3449900030	11-09-2014 00:00:00	423000	4	2.5	2660	5539	2	0	0	3	8	2660	
3818400060	31-10-2014 00:00:00	495000	4	2.5	2460	4862	2	0	0	3	8	2460	
4279200060	30-12-2014 00:00:00	420000	4	2.5	2110	9825	2	0	0	3	8	2110	
5631500505	16-06-2014 00:00:00	576000	4	2.5	2440	28405	2	0	0	3	8	2440	
2115200160	05-06-2014 00:00:00	445000	4	2.5	2340	3784	2	0	0	3	8	2340	
1180500070	24-11-2014 00:00:00	335000	4	2.5	2330	7050	2	0	0	3	8	2330	
6699940140	08-09-2014 00:00:00	352000	4	2.5	2470	5015	2	0	0	3	8	2470	
2641400160	05-12-2014 00:00:00	340000	4	2.5	2380	7850	2	0	0	3	8	2380	

Table: Housing_Data (21,423 rows) Column: id (21,423 distinct values)

The CSV dataset was imported into Microsoft Power BI using the Get Data option.

The following data cleaning operations were performed using Power Query Editor:

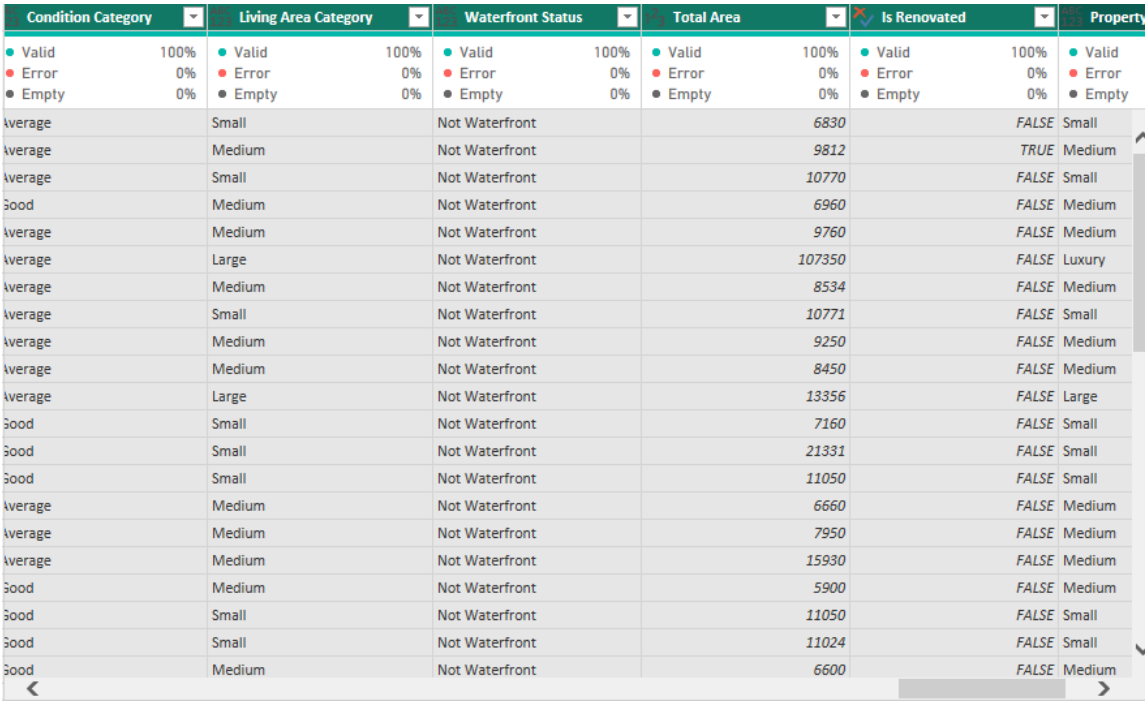
- Checked and corrected the data types of columns.
- Checked important columns for missing and invalid values.
- Created Sale Year and Sale Month from the date column.

- Removed or filtered invalid zero values in bedrooms and bathrooms, where applicable.
- Verified the data before applying the transformations.

5. Data Transformation and Calculated Columns

Several new columns were created to improve the analysis.

Sale Year	Sale Month	Property Age at Sale	Price per SqFt	Price Category
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
2014	10	59	188.0508475	Budget
2014	12	63	209.3385214	Mid-Range
2015	2	82	233.7662338	Budget
2014	12	49	308.1632653	Premium
2015	2	28	303.5714286	Mid-Range
2014	5	13	226.0147601	Premium
2014	6	19	150.1457726	Budget
2015	1	52	275.3301887	Budget
2015	4	55	128.9325843	Budget
2015	3	12	170.8994709	Mid-Range
2015	4	50	186.0955056	Premium
2014	5	72	403.4482759	Mid-Range
2014	5	87	216.7832168	Mid-Range
2014	10	37	291.9708029	Mid-Range
2015	3	115	292.8176796	Mid-Range
2015	1	36	220.3389831	Premium
2014	7	20	208.994709	Mid-Range
2014	5	98	303.125	Mid-Range
2014	12	93	157.5	Budget
2015	4	46	184	Budget
2014	5	67	237.654321	Mid-Range



Condition Category	Living Area Category	Waterfront Status	Total Area	Is Renovated	Property
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
Average	Small	Not Waterfront	6830	FALSE	Small
Average	Medium	Not Waterfront	9812	TRUE	Medium
Average	Small	Not Waterfront	10770	FALSE	Small
Good	Medium	Not Waterfront	6960	FALSE	Medium
Average	Medium	Not Waterfront	9760	FALSE	Medium
Average	Large	Not Waterfront	107350	FALSE	Luxury
Average	Medium	Not Waterfront	8534	FALSE	Medium
Average	Small	Not Waterfront	10771	FALSE	Small
Average	Medium	Not Waterfront	9250	FALSE	Medium
Average	Medium	Not Waterfront	8450	FALSE	Medium
Average	Large	Not Waterfront	13356	FALSE	Large
Good	Small	Not Waterfront	7160	FALSE	Small
Good	Small	Not Waterfront	21331	FALSE	Small
Good	Small	Not Waterfront	11050	FALSE	Small
Average	Medium	Not Waterfront	6660	FALSE	Medium
Average	Medium	Not Waterfront	7950	FALSE	Medium
Average	Medium	Not Waterfront	15930	FALSE	Medium
Good	Medium	Not Waterfront	5900	FALSE	Medium
Good	Small	Not Waterfront	11050	FALSE	Small
Good	Small	Not Waterfront	11024	FALSE	Small
Good	Medium	Not Waterfront	6600	FALSE	Medium

1. **Sale Year:** Extracted the year from the date column.
2. **Sale Month:** Extracted the month from the date column.
3. **Property Age at Sale:** Calculated using Sale Year minus Year Built.
4. **Price per SqFt:** Calculated by dividing Price by Sqft Living.
5. **Price Category:** Categorized houses into Budget, Mid-Range and Premium.
6. **Renovation Status:** Classified properties as Renovated or Not Renovated.
7. **Condition Category:** Classified properties as Poor, Average and Good.

8. **Living Area Category:** Grouped houses into Small, Medium and Large.
9. **Waterfront Status:** Classified properties as Waterfront or Non-Waterfront.
10. **Total Area:** Calculated using Sqft Living plus Sqft Lot.
11. **Is Renovated:** Created a True/False column based on renovation year.
12. **Property Size Bucket:** Categorized properties into Small, Medium, Large and Luxury.

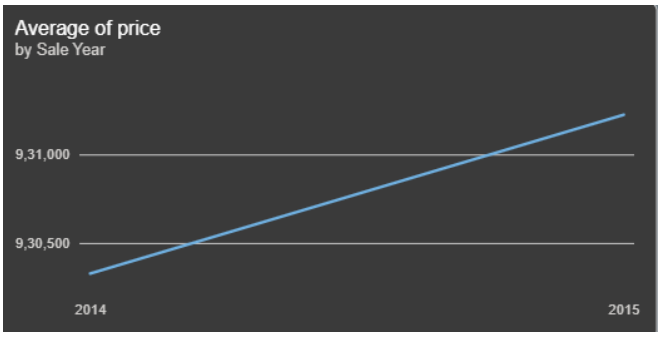
6. Dashboard Development

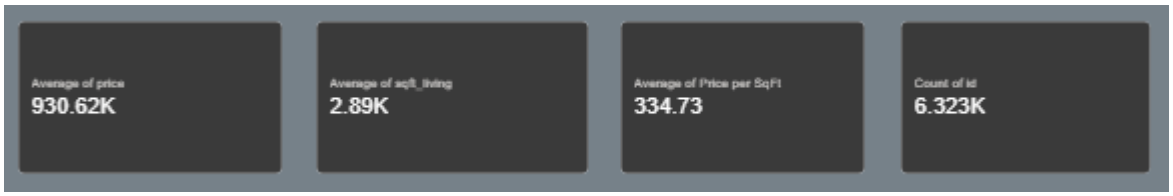
An interactive House Price Analysis Dashboard was developed using Microsoft Power BI.

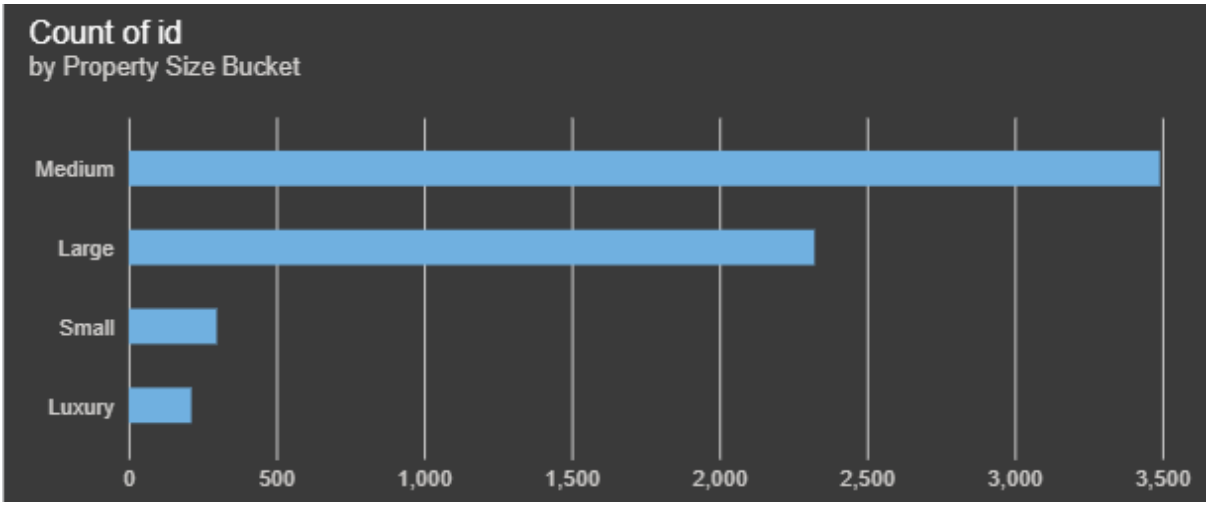
The dashboard includes the following visualizations:

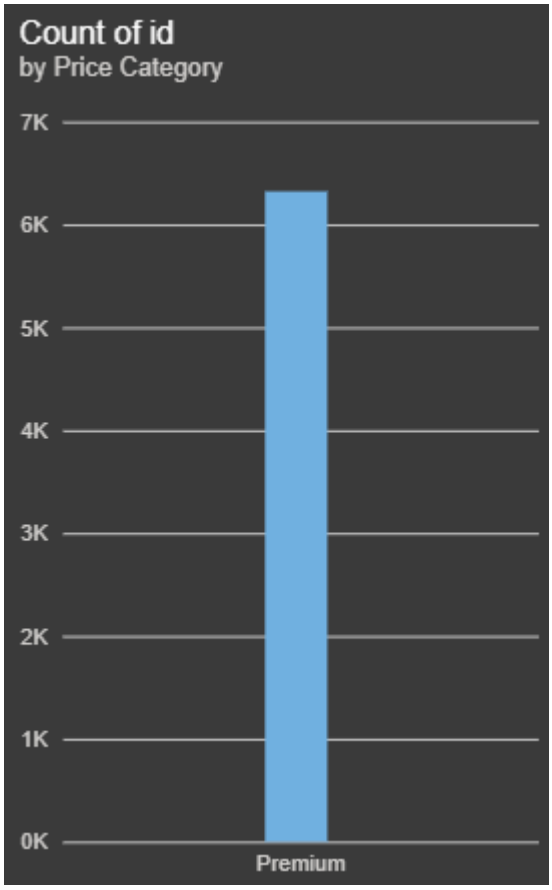
- **Total Properties:** Displays the total number of properties.
- **Average House Price:** Shows the average selling price.
- **Average Living Area:** Displays the average living area.
- **Average Price per SqFt:** Shows the average price per square foot.
- **Average Price by Sale Year:** A line chart used to compare average house prices across years.

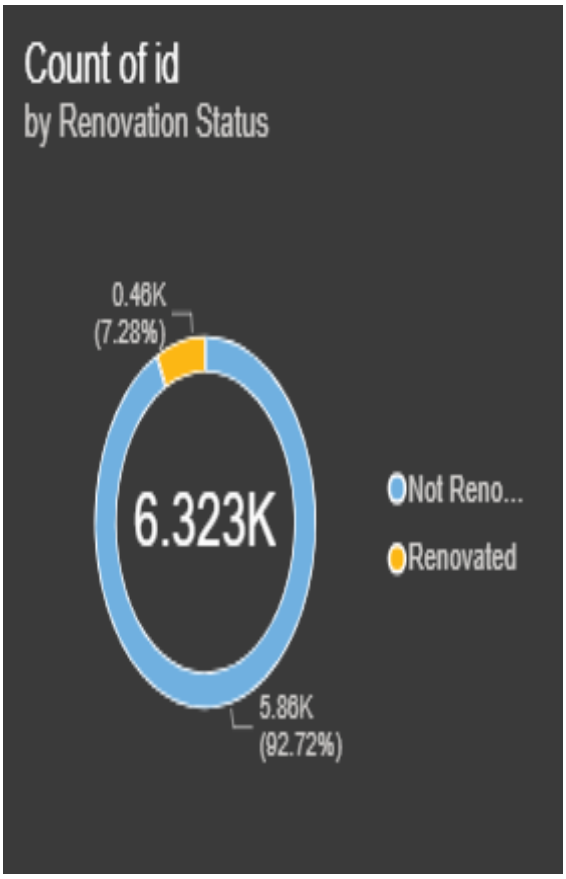
- **Renovation Status:** A donut chart showing renovated and non-renovated properties.
- **Properties by Price Category:** A column chart comparing different price categories.
- **Properties by Size:** A bar chart showing the distribution of properties by size.











7. Interactive Filters

Two slicers were added to make the dashboard interactive:

- **Sale Year:** Allows users to filter properties based on the year of sale.
- **Price Category:** Allows users to filter properties based on their price range.

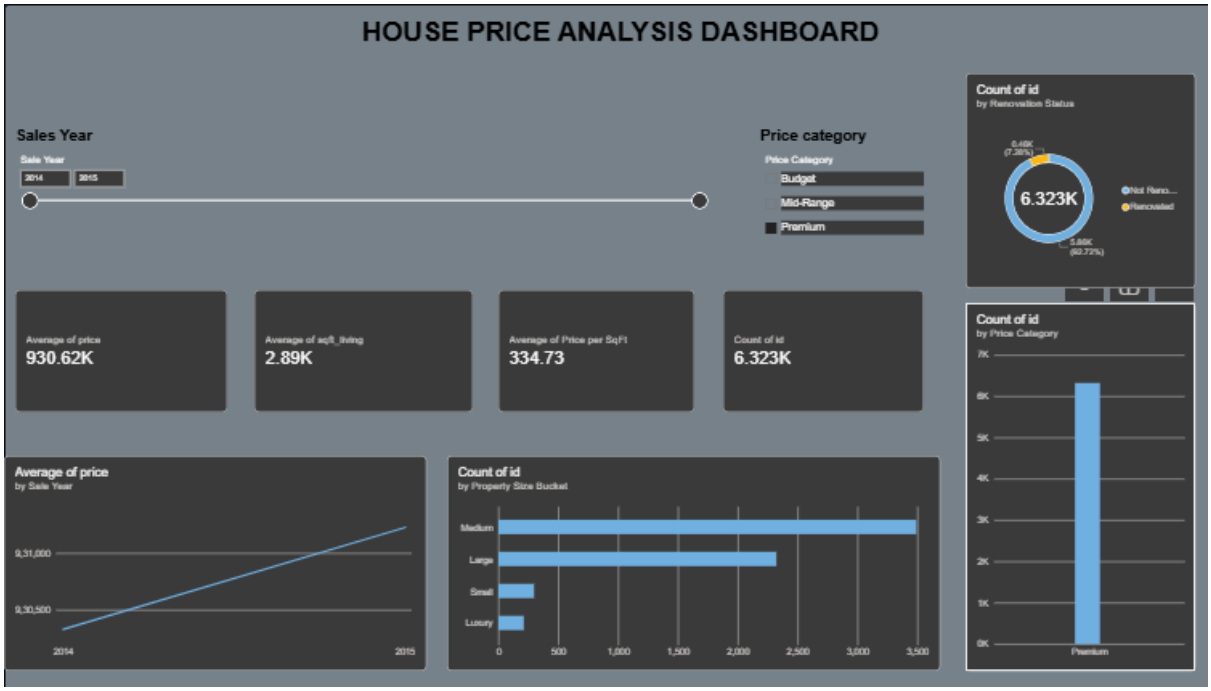
These filters help users explore the data and compare different property categories.



8. Key Observations

The dashboard helps identify the following aspects of the housing dataset:

1. The total number of properties included in the analysis.
2. The average selling price of houses.
3. The distribution of properties across different size categories.
4. The comparison between renovated and non-renovated properties.
5. The changes in average house prices across sale years.



9. Conclusion

The House Price Analysis project demonstrates how Microsoft Power BI can be used for data cleaning, transformation and visualization. The interactive dashboard provides a clear understanding of house prices and property characteristics.

This project helped in learning Power Query, calculated columns, data categorization, KPI cards, charts and slicers. It also improved practical knowledge of data analysis and dashboard development using Power BI.